Effect of a Water Table Aquitard on Drawdown in an Underlying Pumped Aquifer

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A theoretical derivation of the convolution integral produced by N. S. Boulton's originally empirical 'delayed yield' theory shows that it describes the vertical velocity at the base of a water table aquitard having negligible compressibility. The derivation requires that fluid compressibility be neglected and that water be obtained from storage by instantaneous desaturation (or saturation) with a declining (or rising) water table. An expression for the 'delay index' shows that it is a function of hydraulic conductivity, specific yield, and thickness of the aquitard. To evaluate the effect that gradual desaturation may have on flow in a water table aquitard, flow in the unsaturated zone was approximated by an analytical solution of Richards' equation assuming time invariant power law functions for change of both capillary conductivity and moisture content in the vertical. An equation was derived that shows that the length of time for which gradual desaturation could influence flow in the saturated zone is a function of the hydraulic conductivity and specific yield of the aquitard and the thickness of the capillary fringe. To investigate the effect that delayed yield could have on flow in the aquifer, analytical solutions were made for flow to a well pumping at a constant rate in a compressible aquifer overlain by a compressible water table aquitard. Comparison of results that consider delayed yield effects with those that do not suggests that the unsaturated zone has little effect on flow in the aquifer, and comparison of these solutions with numerical solutions that treat the unsaturated zone explicitly conforms with this result.

In the study of groundwater flow involving a water table, three basic approaches are generally taken: variations on the classical free surface theory [e.g., *Boulton*, 1954; *Stallman*, 1965; *Neuman and Witherspoon*, 1971; *Todsen*, 1971], the well-known Dupuit-Forchheimer theory, and variations on the two-phase immiscible flow theory [e.g., *Rubin*, 1968; *Verma and Brutsaert*, 1970; *Freeze*, 1971*a*, *b*; *Cooley*, 1971]. A fourth approach was taken by *Boulton* [1955, 1963, 1970] and *Boulton and Pontin* [1971] to approximate empirically the 'delayed yield' from storage, including water transfer from the unsaturated zone across the water table; this approach was thus intended as an approximation of the third general approach. Boulton has never given a theoretical justification for his method, but it has been noted that his early analytical solution for drawdown around a well pumping at a constant rate in an aquifer affected by delayed yield [*Boulton*, 1963] often described pumping test data from presumed water table aquifers [*Boulton*, 1964; *Prickett*, 1965].

A previous paper by *Cooley* [1971] noted close correspondence between drawdowns generated by *Boulton's* [1963] solution and those produced by a numerical solution for unconfined flow taking the unsaturated zone into account. It was also shown that an analogous correspondence could be obtained for a problem involving a water table aquitard overlying an aquifer being pumped at a constant rate.

These cases involving close agreement of the two approaches lead us to investigate the theoretical justification of the method derived by *Boulton* [1955] and to attempt to formulate a more general analytical approximation of flow from the unsaturated zone into the saturated zone. Consequently, in this paper we first explore the theoretical foundation of Boulton's approximation and show that, to be interpreted deterministically, the approximation does not consider the unsaturated zone. Second, we derive an analytical approximation for vertical flow in the unsaturated zone due to head changes in the saturated zone. Third, we apply this approximation as well as Boulton's and the classical free surface approximations to the aquifer-aquitard problem and compare the results. Finally, we compare the latter results with the numerical solution that takes the unsaturated zone explicitly into account.

THEORETICAL DERIVATION OF BOULTON'S CONVOLUTION INTEGRAL

Boulton [1955, p. 474] assumed that at time t and radius r from the pumped well the part of delayed yield initiated by an increment of drawdown at time τ ($\tau < t$) could be represented by the empirical relationship

$$\delta v_d = -\delta s\, \alpha S_y e^{-\alpha(t-\tau)} \quad (1)$$

where δv_d is the rate of delayed yield at any time t due to drawdown δs in the aquifer during time interval $\delta\tau$ ($\tau < t$); S_y is the specific yield; α is an empirical constant, T^{-1}; and the minus sign was added to indicate that the direction of δv_d is downward, whereas the positive vertical direction is upward. He then assumed that the total rate of delayed yield v_d at radius r and time t could be obtained by integrating over all the delayed yield increments; i.e.,

$$v_d = -\alpha S_y \int_0^t \frac{\partial s}{\partial \tau} e^{-\alpha(t-\tau)}\, d\tau \quad (2)$$

Boulton gave no further theoretical grounds for (1) or (2), the latter of which will be referred to in this paper as Boulton's integral.

To provide insight into the physical basis for the integral, it will be derived in an alternate manner. It will be shown to apply for vertical flow through a porous unit within which the water table lies and for which specific storage S_s is negligibly small. Furthermore, for the interpretation given here its derivation requires that drainage due to water table decline be instantaneous. With use of these ideas the solution is formulated as a unit step increase in drawdown at the bottom boundary of the porous unit at time $t = \tau$ and is then expanded to apply for variable drawdown at the boundary.

In the following analysis, drawdown of the water table s_w is assumed to be much smaller than the thickness of the unit so that solution can be made in the original saturated shape; i.e., deformation of the saturated region due to water table decline can be neglected. For the conditions stipulated in this paragraph and the preceding paragraph the initial value problem can be stated as

$$\begin{aligned}
&\partial^2 s/\partial z^2 = 0 && z_B \le z \le z_w && t > \tau \\
&\frac{\partial s}{\partial z} = -\frac{S_y}{K}\frac{\partial s_w}{\partial t} && z = z_w && t > \tau \\
&s = s_w && z = z_w && t > \tau \\
&s = 1 && z = z_B && t > \tau \\
&s = 0 && z_B \le z \le z_w && t = \tau
\end{aligned} \quad (3)$$

where K is the hydraulic conductivity, z is the vertical dimension (positive upward), z_B is the elevation of the lower boundary of the porous unit, z_w is the elevation of the upper boundary (the water table), and other symbols are as defined previously.

The solution is

$$s = u(t-\tau)\left[1 - \frac{z - z_B}{z_w - z_B} e^{-\alpha(t-\tau)}\right] \quad (4)$$

where $u(t-\tau)$ is the unit step function, equal to 0 for $t \le \tau$ and 1 for $t > \tau$, and $\alpha = K/S_y(z_w - z_B)$. Flow rate across the lower boundary ($z = z_B$) is

$$v = K\, \partial s/\partial z = -u(t-\tau)\alpha S_y e^{-\alpha(t-\tau)} \quad (5)$$

The final integral may be formulated by using a modification of Duhamel's theorem [*Carslaw and Jaeger*, 1959, pp. 30–32] suggested by *Venetis* [1970, p. 54]. If the solution of a linear partial differential equation is of the form $s = s_B(\tau)F(z, t-\tau)$ for the initial (i.e., at $t = \tau$) condition $s = 0$ and the boundary condition $s = s_B(\tau)$, then at any time t for the boundary condition $s = s_B(t)$ the solution is

$$s = \int_0^t \frac{\partial s_B(\tau)}{\partial \tau} F(z, t-\tau)\, d\tau \quad (6)$$

where s_B is the drawdown at the lower boundary and $F(z, t-\tau)$ is the unit step solution. To obtain the flow rate across the bottom boundary, apply Darcy's law to (6) to yield

$$v_d = \int_0^t \frac{\partial s_B(\tau)}{\partial \tau}\left[K\frac{\partial F(z, t-\tau)}{\partial z}\right]\Bigg|_{z=z_B} d\tau \quad (7)$$

Applying (6) and (7) to (4) and (5), we obtain

$$s = s_B - \frac{z - z_B}{z_w - z_B} \int_0^t \left. \frac{\partial s}{\partial \tau} \right|_{z=z_B} e^{-\alpha(t-\tau)}\, d\tau \quad (8)$$

and

$$v_d = -\alpha S_y \int_0^t \left. \frac{\partial s}{\partial \tau} \right|_{z=z_B} e^{-\alpha(t-\tau)}\, d\tau \quad (9)$$

If drawdown at $z = z_B$ is regarded as drawdown in an aquifer being pumped and if (9) is added as a source term to the differential equation describing radial flow in the aquifer, the equation that *Boulton* [1955, p. 474] solved results. Thus, for this case, (2) and (9) are identical. The physical situation described is that of a negligibly compressible aquitard overlying a compressible aquifer. The only difference between the present case and the well-known *Hantush and Jacob* [1955] leaky aquifer problem is that in the present case the upper boundary condition in the aquitard is that of instantaneous release of water from storage at the water table, whereas the upper boundary of Hantush and Jacob's solution is a zero drawdown condition.

Note that the K and S_y appearing in α pertain to the aquitard. Hereafter quantities referring to the aquifer will be labeled with subscript 1, and quantities referring to the aquitard will be labeled with subscript 2. Thus, for the *Boulton* [1955, 1963] solution, α is redesignated for use here as $\alpha_2 = K_2/(S_{y2}b_2)$, where $b_2 = z_w - z_B$.

A critical conclusion is that, as derived here, (9) coupled with the radial flow equation does not allow for delayed yield of the unsaturated zone. Indeed it specifies exactly the opposite, namely, instantaneous yield at the water table. Furthermore, S. P. Neuman (unpublished manuscript, 1972) has suggested that for problems involving purely unconfined flow the effect of vertical flow components on recorded drawdown is incorporated into α_2. This idea was also implied by *Stallman* [1971, pp. 19–23], who described the confusion, in analyzing unconfined flow to a pumping well, that results from close similarities between the solution taking delayed yield into account [*Boulton*, 1963] and the solutions taking vertical flow into account [*Boulton*, 1954; *Stallman*, 1965]. It would appear that, to maintain deterministic meaning, *Boulton*'s [1963] solution is best applied to an aquifer-aquitard system with the restrictions that $S_{s2} \rightarrow 0$ and drainage at the water table can be considered to be instantaneous.

Boulton [1955, p. 473] originally intended his solution to apply only to the aquifer-aquitard system but did not specify the exact nature of the delayed yield mechanism. However, in a later paper [*Boulton*, 1963, p. 470] he specified that delayed yield comes from above the water table, implying that the effect is one of unsaturated flow.

Yotov [1968, 1970] has derived pumping test solutions for a problem involving upward leakage through an incompressible aquitard underlying the pumped aquifer and delayed yield, described by the Boulton integral, affecting the pumped aquifer. *Yotov* [1970, p. 825] illustrated two water table cases to which his solution would apply: (1) leakage through an aquitard overlying the leaky aquifer and containing the water table and (2) purely unconfined flow in the leaky aquifer. On the basis of the derivation and discussion of the Boulton integral given here, applying Yotov's solution to his second case would lead to the same problems as applying it to unconfined flow involving no leakage does.

An interesting result of the explanation of α_2 given here is the interpretation of the inverse relationship of $1/\alpha_2$, which *Boulton* [1963, p. 477] termed the delay index, to the lithology of materials through which drainage takes place given by *Prickett* [1965, p. 12]. If we assume that K_2 will vary more than S_{y2} or b_2, Prickett's relationship gives the expected result that, on the average, the coarser the grain size, the more permeable the material. In making this interpretation we must assume additionally that, because materials that are usually finer grained than the specified aquifers overlie the aquifers, as is indicated by the driller's logs [*Prickett*, 1965, p. 14], these materials are reacting as slightly compressible aquitards. If this assumption is correct, the effects of unsaturated flow may be neglected because, as is shown further on, the effects of unsaturated flow on drawdown in a well that is open to the aquifer in an aquifer-aquitard system are not significant.

ALTERNATE FORMULATION OF YIELD FROM THE UNSATURATED ZONE

A deterministic water table boundary condition would aid in the study of delayed yield phenomena. For a problem of the type illustrated in Figure 1, in which flow in the unsaturated zone can be assumed to be nearly vertical, a water table boundary condition similar in some respects to Boulton's integral may be derived by using an approximation of the theory of unsaturated flow.

If compressibility of the aquitard and water is neglected in the unsaturated zone, the equation for vertical unsaturated flow is a form of Richards' equation [*Richards,* 1931, p. 324], which may be written in terms of drawdown as

$$\frac{\partial}{\partial z}\left[K_c(\theta_r)\frac{\partial s_c}{\partial z}\right] = \frac{d\theta_r}{dp_h}\frac{\partial s_c}{\partial t} \qquad (10)$$

where p_h is the water pressure head, $K_c(\theta_r)$ is the capillary conductivity in the aquitard as a function of θ_r, θ_r is the drainable water content per unit volume and is equal to the total water content minus the residual water content, and s_c is the drawdown in the unsaturated zone.

Equation 10 must be linearized to be solved analytically. Linearization may be accomplished by using the idea that the solution may be made with the original shape of the unsaturated zone. If we assume initially static conditions, (10) may then be rewritten as

$$\frac{\partial}{\partial z}\left(K_c(z)\frac{\partial s_c}{\partial z}\right) = -\frac{d\theta_r}{dz}\frac{\partial s_c}{\partial t} \qquad (11)$$

In this study we will assume that the two functions of z (Figure 2) may be approximated by

$$K_c(z) = (K_2/L_c^n)(b_T + L_c - z)^n \qquad (12)$$

and

$$\theta_r = (S_{y2}/L_c^n)(b_T + L_c - z)^n \qquad (13)$$

where n is a positive integer, $b_T = b_1 + b_2 + L_2$, L_2 is the thickness of the saturated or nearly saturated zone above the water table, and L_c is the thickness of the unsaturated zone defined between $z = b_T$ and the point where $\theta_r \to 0$.

Combining (12) and (13) demonstrates that

$$K_c(\theta_r) = (K_2/S_{y2})\theta_r \qquad (14)$$

is a linear approximation of the actual function.

Substituting (12) and (13) into (11) yields

$$(b_T + L_c - z)\frac{\partial^2 s_c}{\partial z^2} - n\frac{\partial s_c}{\partial z} = \frac{1}{D}\frac{\partial s_c}{\partial t} \qquad (15)$$

where $D = K_2/nS_{y2}$. The problem is formulated as a unit step function of drawdown at the lower boundary $z_B = b_T$ at time τ so that (6) and (7) can be used to obtain the final solution. Boundary and initial conditions are

$$\begin{array}{lll} s_c = 1 & z = b_T & t > \tau \\ v = 0 & z = b_T + L_c & t > \tau \\ s_c = 0 & b_T \le z \le b_T + L_c & t = \tau \end{array} \qquad (16)$$

where v is the discharge per unit area. The unit step solution is obtained by using the separation of variables method (Appendix 1) and is

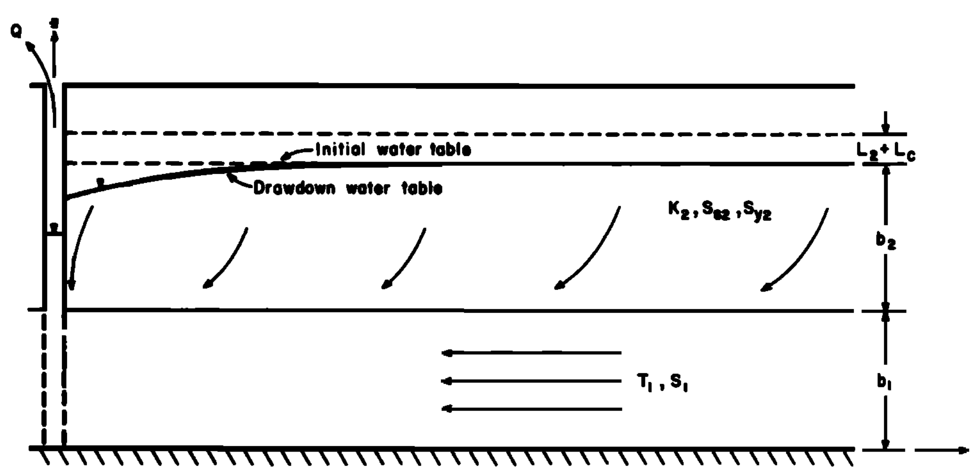


Fig. 1. Diagram and notation used for the problem of flow to a well in an aquifer overlain by a water table aquitard.

$$s_c = 1 - \frac{1}{L_c^{1/2}} \left(\frac{b_T + L_c - z}{L_c} \right)^{(1-n)/2}$$

$$\cdot \sum_{m=1}^{\infty} \frac{J_{1-n}[2k_m(b_T + L_c - z)^{1/2}]}{k_m J_{2-n}(2k_m L_c^{1/2})} e^{-k_m^2 D(t-\tau)} \quad (17)$$

where J_l is the Bessel function of the first kind and order l and the k_m are chosen such that $J_{1-n}(2k_m L_c^{1/2}) = 0$.

The final solution, obtained by applying (6) to (17), is therefore

$$s_c = s_c|_{z=b_T} - \frac{1}{L_c^{1/2}} \left(\frac{b_T + L_c - z}{L_c} \right)^{(1-n)/2}$$

$$\cdot \sum_{m=1}^{\infty} \frac{J_{1-n}[2k_m(b_T + L_c - z)^{1/2}]}{k_m J_{2-n}(2k_m L_c^{1/2})}$$

$$\cdot \int_0^t \left. \frac{\partial s_c}{\partial \tau} \right|_{z=b_T} e^{-k_m^2 D(t-\tau)} \, d\tau \quad (18)$$

Outflow is found by combining (7) and (18):

$$v_d = -\alpha_2' S_{y2} \int_0^t \left. \frac{\partial s_c}{\partial \tau} \right|_{z=b_T}$$

$$\cdot \sum_{m=1}^{\infty} \exp \left[-\frac{R_m^2}{4n} \alpha_2'(t - \tau) \right] d\tau \quad (19)$$

where $\alpha_2' = K_2/S_{y2}L_c$ and $R_m = 2k_m L_c^{1/2}$.

It is of value to determine the length of time that the values of n and L_c influence yield from the unsaturated zone. This may be accomplished most easily by taking the Laplace transform of the gradient of (18) at $z = b_T$ (see Appendix 2) and then examining the asymptotic behavior of the result at large values of time:

$$\left. \frac{\partial s_c'}{\partial z} \right|_{z=b_T} = -s_c' \Big|_{z=b_T} \left(\frac{p}{L_c D} \right)^{1/2}$$

$$\cdot \frac{I_n[2(pL_c/D)^{1/2}]}{I_{n-1}[2(pL_c/D)^{1/2}]} \quad (20)$$

where s_c' is the Laplace transform of drawdown s_c, p is the Laplace transform variable, and I_l is the modified Bessel function of the first kind and order l. As p becomes large, t becomes small, and vice versa [*Hantush*, 1960, p. 3722]. For $p \leq nD/16L_c$ ($t \geq 16L_c/nD$) equation 20 can be approximated by using the asymptotic relationship

$$I_l(x) = (1/l!)(x/2)^l \quad (21)$$

[*Boas*, 1966, p. 573], which results in

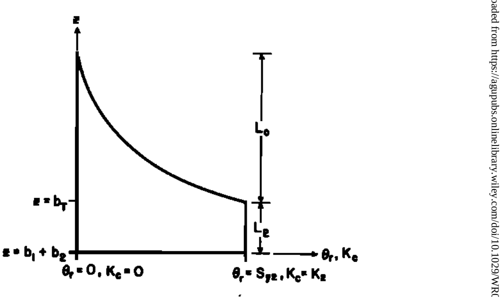


Fig. 2. Notation used for the assumed relationships of drainable water content θ_r and capillary conductivity K_c to elevation above the aquifer base z.

$$\left. \frac{\partial s_c'}{\partial z} \right|_{z=b_T} = -\frac{S_{y2}}{K_2} p s_c' |_{z=b_T} \quad (22)$$

Equation 22 can be inverted directly to give

$$\left. \frac{\partial s_c}{\partial z} \right|_{z=b_T} = -\frac{S_{y2}}{K_2} \left. \frac{\partial s_c}{\partial t} \right|_{z=b_T} \quad (23)$$

which is simply the 'no delay' free surface condition. Thus under the assumptions used to derive (19) the unsaturated zone has little or no influence on drawdown after $t = 16L_c/nD$. This conclusion will be explored further in a later section of this paper.

SOLUTIONS OF THE AQUIFER-AQUITARD PROBLEM

In this section three solutions to the problem illustrated schematically in Figure 1 will be compared, the objects being to present a solution to the aquifer–water table aquitard problem that incorporates the effects of storage in the aquitard and to determine the sensitivity of the solution to different ways of describing the water table. The solutions will differ only because of the following boundary conditions used to describe the water table.

Boulton's integral. For application to the present problem we assume that the integral describes the vertical drainage of a saturated or nearly saturated zone of thickness L_2 above the water table. Physically, this case incorporates the assumption made by *Freeze* [1971*a*, p. 350] that, for flow at less than atmospheric pressures, porous medium compressibility is essentially 0. In addition, it neglects flow in the unsaturated zone and assumes that the effect of fluid compressibility in the zone defined by L_2 is negligible.

No delay. The upper limit of the compressible aquitard includes the saturated or nearly saturated zone and has the no delay boundary condition. This case differs from Boulton's integral only in that the zone defined by L_2 now has a specific storage equal to S_{s2}.

Approximation for the unsaturated zone. This condition was developed in the preceding section and is described by (19). However, because of the difficulty of obtaining the complete solution incorporating (19), the asymptotic equation (23) is used in this paper. Using (23) simply means that the results of the no delay case are used before and after the time period when the effects of delayed yield could be present. The effects of delayed yield on the solution in the aquifer are analyzed numerically in the next section.

In the mathematical formulation of the problem, flow in the aquitard is assumed to be vertical, and flow in the aquifer is assumed to be horizontal. These two assumptions were made by *Hantush* [1960] to solve a problem similar to that considered here but without the influence of the water table. For this case, they were recently shown to yield errors in drawdown in the aquifer generally of <5% for ratios of aquifer to aquitard conductivity as low as 3 [*Neuman and Witherspoon*, 1969, p. 134]. The assumptions led to much smaller errors for larger ratios.

For the Boulton integral boundary condition, the initial value problem may be stated for the aquitard as

$$\frac{\partial^2 s_2}{\partial z^2} = \frac{S_{s2}}{K_2}\frac{\partial s_2}{\partial t}$$

$$0 < r < \infty \qquad b_1 \le z \le b_1 + b_2 \qquad t > 0$$

$$\frac{\partial s_2}{\partial z} = -\frac{\alpha_2 S_{y2}}{K_2}\int_0^t \frac{\partial s_2}{\partial \tau} e^{-\alpha_2(t-\tau)}\, d\tau$$

$$0 < r < \infty \qquad z = b_1 + b_2 \qquad t > 0 \tag{24}$$

$$s_2 = s_1$$

$$0 < r < \infty \qquad z = b_1 \qquad t > 0$$

$$s_2 = 0$$

$$0 < r < \infty \qquad b_1 < z < b_1 + b_2 \qquad t = 0$$

where $\alpha_2 = K_2/S_{y2}L_2$. In the aquifer

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial s_1}{\partial r}\right) + \frac{K_2}{T_1}\frac{\partial s_2}{\partial z}\bigg|_{z=b_1} = \frac{S_1}{T_1}\frac{\partial s_1}{\partial t}$$

$$0 < r < \infty \qquad t > 0$$

$$\lim_{r\to 0} r\frac{\partial s_1}{\partial r} = -\frac{Q}{2\pi T_1} \qquad t > 0 \tag{25}$$

$$\lim_{r\to\infty} s_1 = 0 \qquad t > 0$$

$$s_1 = 0 \qquad 0 < r < \infty \qquad t = 0$$

This problem is the same as that studied by *Hantush* [1960] except for the upper boundary condition. He considered both a zero drawdown and an impermeable upper boundary, whereas in this problem the upper boundary condition uses the Boulton integral.

The complete solution to the problem is divided into early and late time segments. The solution for early time ($t < b_2S_2/10K_2$) for the aquifer is the same as that given by *Hantush* [1960, p. 3716] for early time if his underlying aquitard is assumed to be impermeable. It is

$$s_1 = (Q/4\pi T_1)H_a(u_a, \beta) \tag{26}$$

where

$$H_a(u_a, \beta) = \int_{u_a}^{\infty}\frac{e^{-y}}{y}\operatorname{erfc}\left(\frac{\beta(u_a)^{1/2}}{[y(y-u_a)]^{1/2}}\right)dy$$

$$\beta = \frac{r}{4}\lambda \qquad \lambda = \left(\frac{K_2S_{s2}}{T_1S_1}\right)^{1/2} \qquad u_a = \frac{r^2S_1}{4T_1t}$$

For late time ($t \ge 10b_2S_2/K_2$) the solution is similar to *Boulton*'s [1963, p. 479] complete solution and can be written in nondimensionalized form as

$$s_1 = \frac{Q}{4\pi T_1}\int_{\mu_{D0}}^{\infty}\left[1 - e^{-\mu_{D1}}\left(\cosh \mu_{D2} + \frac{(r/B)^2\eta - 4\nu^2y^2}{4\mu_{D2}\nu^2u_y}\sinh \mu_{D2}\right)\right] \cdot J_0\left[\left(\frac{8y^2}{\eta} - \frac{(r/B)^2}{\nu^2}\right)^{1/2}\right]\frac{16\nu^2y\, dy}{8\nu^2y^2 - \eta(r/B)^2} \tag{27}$$

where

$$\mu_{D1} = y^2/u_y$$

$$\mu_{D2} = \frac{1}{4u_y\nu^2} \cdot \{16\nu^4y^4 - (r/B)^2[8\nu^2y^2 - \eta(r/B)^2]\}^{1/2}$$

$$\mu_{D3} = [\eta(r/B)^2/8\nu^2]^{1/2}$$

$$B = \left(\frac{T_1}{\alpha_2 S_{y2}} + \frac{T_1}{K_2/b_2}\right)^{1/2} = \left[\frac{T_1(L_2 + b_2)}{K_2}\right]^{1/2}$$

$$u_y = (S_1 + S_2 + S_{y2})r^2/4T_1 t$$

$$\eta = \frac{S_1 + S_2 + S_{y2}}{S_1 + K_2 S_2/(K_2 + \alpha_2 S_{y2} b_2)} = \frac{S_1 + S_2 + S_{y2}}{S_1 + S_2/(1 + b_2/L_2)}$$

$$\nu^2 = S_{y2}/(S_1 + S_2 + S_{y2})$$

An outline of the derivation of (26) and (27) can be found in Appendix 3.

It was found that a simplified solution for the case in which $\eta \rightarrow \infty$ can be used for a wide range of finite values (this finding is similar to what was found by *Boulton* [1963, pp. 471, 473–475]). The solution for this case in nondimensionalized form is

$$s_1 = (Q/4\pi T_1)H_y(u_y, r/B) \qquad (28)$$

where

$$H_y\left(u_y, \frac{r}{B}\right) = \int_0^\infty 2J_0(2y)\left[1 - \frac{(r/B)^2}{(r/B)^2 + 4y^2} \cdot \exp\left(\frac{-(r/B)^2 y^2}{u_y[(r/B)^2 + 4y^2]}\right)\right]\frac{dy}{y}$$

$$u_y = r^2 S_{y2}/4T_1 t$$

and other symbols are as defined above.

The solution using the no delay boundary condition is obtained by modifying (24). The upper boundary is placed at $b_T = b_1 + b_2 + L_2$ instead of $b_1 + b_2$, and the upper boundary condition is

$$\partial s_2/\partial z = -S_{y2}/K_2\, \partial s_2/\partial t \qquad (29)$$

$$0 < r < \infty \qquad z = b_T \qquad t > 0$$

Equation 26 describes the early time solution, and (27) describes the complete late time solution if the following redefinitions are made:

$$S_2 = S_{s2}(b_2 + L_2)$$

and

$$\eta = (S_1 + S_2 + S_{y2})/S_1$$

Equation 28 describes the simplified late time solution. Both late time solutions may be obtained from (27) and (28), respectively, by allowing $\alpha_2 \rightarrow \infty$ as $L_2 \rightarrow 0$ and then redesignating the aquitard thickness as $b_2 + L_2$ instead of b_2.

The final solution of the three cases considered here is identical to that for no delay except that possible 'delayed yield' effects theoretically limit application of the late time solution to $t \geq 16S_{y2}L_c/K_2$.

The principal conclusion to be reached from discussion of the three solutions is that they are almost identical, and, as $L_2 \rightarrow 0$, they are completely identical except for the added time restriction for the third solution. The first two solutions neglect the effect of the unsaturated zone, but the third defines a time span during which its effects could be felt. The effect that the unsaturated zone has on drawdown in the aquifer must appear between $t = b_2 S_2/10K_2$ and $t = 16S_{y2}L_c/K_2$ and then cannot be too great in order to preserve continuity of the complete drawdown curve. Thus, if the approximations made are valid, the near correspondence of the solutions indicates that the unsaturated zone has very little effect on drawdown in the aquifer. A further conclusion to be drawn from the correspondence of the solutions is that, for the normal case in which $L_2 \ll b_2$, neglecting compressibility in the zone defined by L_2 also has little effect on drawdown in the aquifer.

Asymptotic early and late time solutions express the basic physical nature of the flow system. For some time after the initiation of pumping, a negligible amount of water is released from storage as a result of desaturation; nearly all the water being pumped comes from elastic storage in the aquitard and aquifer. During this time span, for all practical purposes the drawdown has not affected the water table. The solution to this problem thus conforms to the early time solution given by *Hantush* [1960, p. 3716] if $S_{s2} > 0$ or to that given by *Hantush and Jacob* [1955] if $S_{s2} \rightarrow 0$. For some later time span (i.e., $t > 10b_2S_2/K_2$), water is withdrawn from storage principally as a result of desaturation due to drawdown of the water table, and withdrawal of water from elastic storage is negligible. Solution during this time is approximated by the late time curves derived here if $S_{s2} > 0$ or by *Boulton*'s [1955, 1963] solution if $S_{s2} \rightarrow 0$.

COMPARISON WITH NUMERICAL SOLUTIONS

Testing the concepts developed in the previous sections was accomplished by using approximately the same procedures as those described by *Cooley* [1971] for numerically analyzing unconfined or semiconfined flow to a well. Cooley's model makes none of the assumptions employed to linearize the equations or reduce the dimensionality of the flow adopted for the present study. Instead it treats the problem directly in terms of two-dimensional (radial and vertical) slightly compressible flow in a variably saturated, slightly compressible porous medium. Details of assumptions made and general methods used are given by *Cooley* [1971].

Two problems, which differ only in properties of the unsaturated zone, were analyzed numerically. The following equations were used to relate pressure head p_h and relative conductivity $K_r = K_c/K_2$ to drainable water content θ_r:

$$\theta_r = S_{y2}A/[(-p_h)^c + A] \tag{30}$$

and

$$K_r = (\theta_r/S_{y2})^d \tag{31}$$

Equations 30 and 31 are equivalent to (12) and (13) by *Cooley* [1971, p. 1610].

Properties of the unsaturated zone were changed by varying c and A. For the first of the two test problems, $c = 4$ and $A = 1000$, these values corresponding to porous material with high moisture retention (Figure 3). For the second problem, $c = 4$ and $A = 0.1$, these values corresponding to porous material with low moisture retention (Figure 3). Although

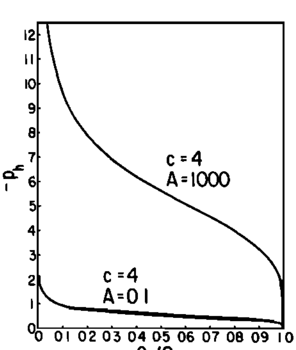


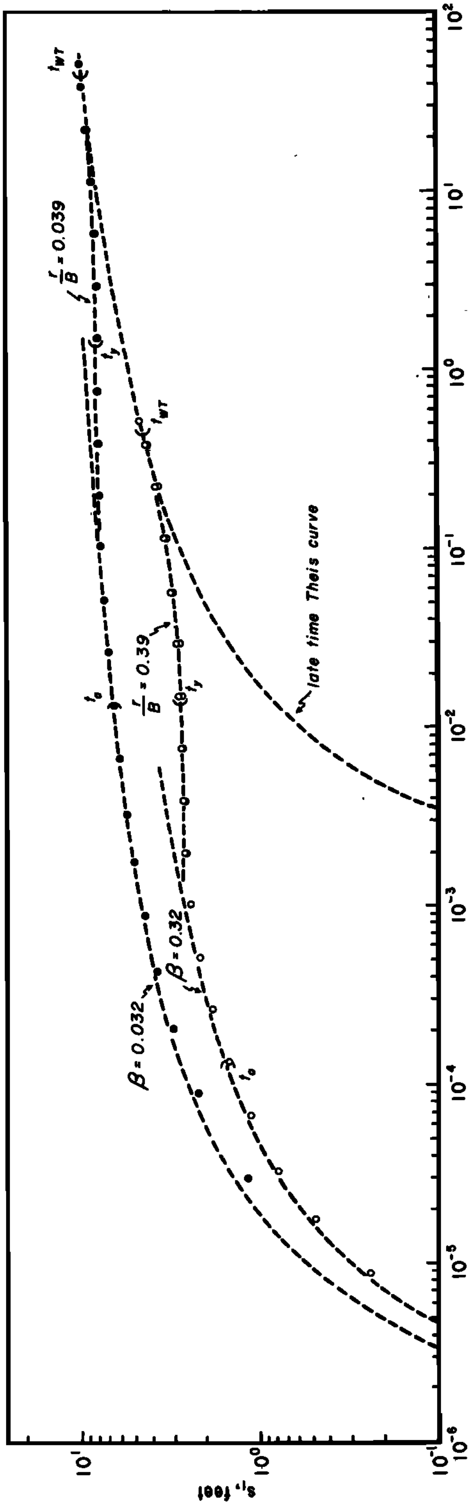
Fig. 3. Relationships of drainable water content θ_r to water pressure head p_h, used for the numerical solutions.

the second soil moisture retention curve is somewhat unrealistic when it is applied to an aquitard, it fit one of the purposes of making the numerical solutions, which was to examine the variability of drawdown due to changes in soil moisture retention.

Exponent d usually lies between the narrow limits of 3.5 and 4.0 [*Brooks and Corey*, 1966, p. 78]; however, d used for the analytical solution has the value of 1 because (14) is equivalent to (31) when $d = 1$. Previous numerical experiments have indicated some degree of insensitivity of head distribution below the water table to changes in d. Therefore we decided to compare the analytical and numerical solutions by using the physically realistic value of $d = 4$ for the numerical model.

Assumed parameters for the aquifer and aquitard are listed in Figure 4. For both problems 40 radii of node points were spaced geometrically from the well radius $r_w = 0.5$ feet to an external radius of 18,221 feet. Rows of node points were spaced at 10-foot intervals at 0- to 50-foot elevation and at 2.5-foot intervals at 50- to 120-foot elevation for the first problem. For the second problem the row spacing was decreased to approximately 0.8 feet at 92.5- to 100-foot elevation and to 0.2 feet at 100-foot elevation to the last row at 101.4-foot elevation. The initial water table was placed at 100-foot elevation in both solutions. Time step sizes were generated by multiplying the previous time increment by 1.4, the initial increment being 0.005 min. Forty-four time steps resulted in a total pumping period of 20,270 min.

The results of the test solutions are shown in Figure 4, where $t_a = b_2S_2/10K_2$, $t_y = 10b_2S_2/K_2$, and $t_{WT} = 16S_{y2}L_c/K_2$ ($L_c = 10$ feet from the first problem). The analytical solution used was the no delay condition, which was computed by using an approximate average value of $L_2 = 1$ foot. As was predicted by the approximate theory, both numerical solutions are very nearly equal to the analytical solution, and the difference between the two numerical solutions is very small, variation occurring only between the limits t_a and t_{WT}. The magnitude of the small difference between numerical results cannot be investigated exactly by using the approximate analytical solution because it does not apply between t_a and t_{WT}. However, we can investigate the part of the difference due to the actual



range in two values of L_2. For the first problem, $L_2 \simeq 3$ feet, $r/B = 0.0389$ at $r = 20$ feet, and $r/B = 0.3885$ at $r = 200$ feet. For the second problem, $L_2 \simeq 0.3$ feet, $r/B = 0.0398$ at $r = 20$ feet, and $r/B = 0.3980$ at $r = 200$ feet. Drawdown differences for these values of r/B in the range t_y–t_{WT} are of the same order of magnitude as those shown in Figure 4. Therefore values of L_c, L_2, and by inference n had little effect on drawdown in the aquifer for the problems examined. Correspondence of the numerical and analytical solutions also tends to reinforce the idea of insensitivity of the solution to values of d.

Note that deviation of the numerical results from the analytical curve at early time can be shown to be largely the result of numerical discretization error [*Cooley*, 1971, p. 1621].

CONCLUSIONS

The following conclusions were drawn from the results of this study:

1. *Boulton*'s [1955] convolution integral, which was originally empirically derived, can be theoretically derived to describe vertical velocity at the bottom of a water table aquitard having negligible compressibility. For the derivation, fluid compressibility is also neglected, and water must be obtained from storage by instantaneous desaturation during water table drawdown. With this interpretation Boulton's empirical constant α (redesignated α_2 here to indicate that it is a property of the aquitard) is defined by $\alpha_2 = K_2/S_{y2}b_2$, where the variables have been defined earlier. Data given by *Prickett* [1965, p. 12] showing the inverse correlation between the grain size of materials through which drainage takes place and the delay index $(1/\alpha_2)$ conform qualitatively with the above expression for α_2.

In the introduction it was noted that pumping test data from assumed unconfined aquifers

Fig. 4. Relationship of t/r^2 to drawdown s_1 for the problem of flow to a well in an aquifer overlain by a water table aquitard. The numerical solutions are indicated by the circles, and the analytical solution is indicated by the dashed lines. The symbols t_a, t_y, and t_{WT} are defined in the text, $\beta = (r/4)(K_2S_{s2}/T_1S_1)^{1/2}$, and $B = [T_1(b_2 + L_2)/K_2]^{1/2}$, where $T_1 = 50{,}000$ gal/day/ft, $S_1 = 0.0001$, $K_2 = 10$ gal/day/ft², $S_2 = 0.001$, $S_{y2} = 0.1$, $b_1 = 50$ ft, $b_2 = 50$ ft, $r_w = 0.5$ ft, and $Q = 500$ gal/min.

often fit the *Boulton* [1963] solution closely. On the basis of the results of this study, one may speculatively cite three possible reasons for this close fit. One is that systems such as many of those given in *Prickett* [1965, p. 14] may in reality be semiconfined by a water table aquitard. The second is that the similarity in form between the two-dimensional (radial and vertical) no delay unconfined aquifer solution that includes compressibility (S. P. Neuman, unpublished manuscript, 1972) and *Boulton*'s [1963] solution allows calculation of an effective α that incorporates the effects of vertical flow (S. P. Neuman, unpublished manuscript, 1972). Finally, it seems probable that other cases involving the effects of transition from the dominance of one storage coefficient to the dominance of another would also yield curves with the typical S shape.

2. An analytical approximation for vertical flow in the unsaturated zone allows for power law variation of both capillary conductivity and moisture content in the vertical. The necessary linearization of Richards' equation was accomplished by assuming that the two curves retain their initial shape throughout the unsteady flow period. By use of these approximations, it was found that the unsaturated zone could influence flow in the saturated zone for a time period up to $t_{WT} = 16S_{y2}L_c/K_2$.

3. Boulton's convolution integral describing vertical incompressible flow through a saturated or nearly saturated capillary fringe and the classical free surface condition were used as water table boundary conditions to solve the problem of a well pumping in an aquifer underlying a water table aquitard. The two solutions resulting from the two boundary conditions are very nearly the same. However, these two boundary conditions do not incorporate the unsaturated zone, and conductivity variation in the unsaturated zone could cause small variations in drawdown in the aquifer between the times $t_a = b_2S_2/10K_2$ and t_{WT}.

4. The analytical solution for the aquifer, which assumes no delayed yield effects, conforms to a numerical solution that includes the unsaturated zone to within the limits of numerical error. Variation of parameters describing soil moisture retention and conductivity caused only a small variation in drawdown in the aquifer between times t_a and t_{WT}. Small variation in drawdown was implied by the analytical solution in order to maintain continuity of the drawdown curve.

APPENDIX 1: DERIVATION OF (17)

A change of variables in (15) and (16) of the text from z to $z' = b_T + L_c - z$ results in the following transformed initial value problem:

$$z' \frac{\partial^2 s_c}{\partial z'^2} + n \frac{\partial s_c}{\partial z'} = \frac{1}{D} \frac{\partial s_c}{\partial t} \tag{A1}$$

$$\begin{aligned} s_c &= 1 & z' &= L_c & t &> \tau \\ v &= 0 & z' &= 0 & t &> \tau \\ s_c &= 0 & L_c \geq z' &\geq 0 & t &= \tau \end{aligned} \tag{A2}$$

The solution is obtained by separating variables with $s_c(z', t - \tau) = Z'(z')T(t - \tau)$ and $-k^2$ as the separation constant, which yields

$$1/DT(t - \tau)\ \partial T(t - \tau)/\partial t = -k^2 \tag{A3}$$

and

$$\frac{\partial^2 Z'(z')}{\partial z'^2} + \frac{n}{z'} \frac{\partial Z'(z')}{\partial z'} + \frac{Z'(z')k^2}{z'} = 0 \tag{A4}$$

Solving (A3) by inspection and (A4) by using the generalized Bessel equation [*Boas*, 1966, p. 565] and recalling that a complete set of functions is required to give the solution for s_c, we obtain the following equation:

$$s_c = 1 - z'^{(1-n)/2} \cdot \sum_{m=1}^{\infty} A_m J_{1-n}(2k_m z'^{1/2}) e^{-k_m^2 D(t-\tau)} \tag{A5}$$

where use has been made of the fact that both drawdown and hydraulic gradient at $z' \to 0$ must remain finite while K_2 and thus v approach 0.

The first boundary condition determines the values of the k_m as

$$k_m = R_m/2L_c^{1/2} \tag{A6}$$

where the R_m are the roots of

$$J_{1-n}(R_m) = 0 \tag{A7}$$

The initial condition fixes the A_m as coefficients of the Fourier-Bessel series in (A5) on evaluating

$$\frac{1}{2^{n-1}}\int_0^{2L_c^{1/2}} x^n J_{n-1}(k_m x)\, dx = (-1)^{n-1} A_m \cdot \int_0^{2L_c^{1/2}} x J_{n-1}^2(k_m x)\, dx \qquad \text{(A8)}$$

[*Bland*, 1961, pp. 43–47], where $x = 2z'^{1/2}$, and solving for A_m, which yields

$$A_m = L_c^{(n-2)/2}/(-1)^n k_m J_{n-2}(2k_m L_c^{1/2})$$
$$= L_c^{(n-2)/2}/k_m J_{2-n}(2k_m L_c^{1/2}) \qquad \text{(A9)}$$

Finally, combining (A5) and (A9) and substituting $z' = b_T + L_c - z$ give (17).

APPENDIX 2: DERIVATION OF (20)

The most efficient means of finding the Laplace transform of the gradient of (18) is to Laplace-transform the original problem as given in (A1) and (A2) with modification of the boundary condition at $z' = L_c$ to include variable s_c. Differentiation may then be accomplished directly in the transformed space. The Laplace transform of (A1) is

$$z' \frac{\partial^2 s_c'(z', p)}{\partial z'^2} + n \frac{\partial s_c'(z', p)}{\partial z'} - \frac{p}{D} s_c'(z', p) = 0 \qquad \text{(A10)}$$

where the prime with s_c denotes that it has been Laplace transformed. Equation A10 may then be solved by using the generalized Bessel equation [*Boas*, 1966, p. 565], which gives

$$s_c'(z', p) = A z'^{(1-n)/2} J_{1-n}[2i(pz'/D)^{1/2}] \qquad \text{(A11)}$$

where $i = (-1)^{1/2}$.

Coefficient A is determined by using the idea that at $z' = L_c$ the value of s_c' is known. Substituting the value of A so obtained in (A11) yields, in terms of modified Bessel functions of the first kind,

$$s_c'(z', p) = s_c'(L_c, p)\left(\frac{z'}{L_c}\right)^{(1-n)/2} \cdot \frac{I_{n-1}[2(pz'/D)^{1/2}]}{I_{n-1}[2(pL_c/D)^{1/2}]} \qquad \text{(A12)}$$

where use has been made of the identity $I_{-m}(x) = I_m(x)$ for a positive integer m [*Bland*, 1961, p. 41]. Replacing z' with $b_T + L_c - z$ and evaluating $\partial s_c'/\partial z|_{z=b_T}$ yield (20).

APPENDIX 3: DERIVATION OF (26) AND (27)

Equations 24 and 25 outline the boundary value problem, which is to be solved by using a combination of Hankel and Laplace transform techniques. The Laplace transform is defined as usual [*Boas*, 1966, p. 592], whereas the Hankel transform is defined as

$$H[F(r)] = \int_0^\infty F(r) J_0(ar) r\, dr = F'(a) \qquad \text{(A13)}$$

[e.g., *Hantush*, 1960, p. 3720], where a is the transform variable. In these equations a prime with the function indicates that it has been either Laplace or Hankel transformed, and a double prime indicates that the function has been doubly transformed. Laplace and Hankel transforming (24) yields

$$\frac{\partial^2 s_2''(a, z, p)}{\partial z^2} - \frac{p}{\nu_2} s_2''(a, z, p) = 0$$

$$\left.\frac{\partial s_2''(a, z, p)}{\partial z}\right|_{z=b_1+b_2} = -\gamma_2 \frac{p s_2''(a, b_1 + b_2, p)}{p + \alpha_2} \qquad \text{(A14)}$$

$$s_2''(a, b_1, p) = s_1''(a, p)$$

where $\nu_2 = K_2/S_{s2}$, $\gamma_2 = \alpha_2 S_{y2}/K_2$, and the initial condition has been used. The solution of (A14) is

$$s_2''(a, z, p) = \frac{s_1''(a, p) \cosh [(p/\nu_2)^{1/2}(b_1 + b_2 - z)]}{\cosh [(p/\nu_2)^{1/2} b_2]} + \frac{s_2''(a, b_1 + b_2, p)(\nu_2 p)^{1/2}\gamma_2 \sinh [(p/\nu_2)^{1/2}(b_1 + b_2 - z)]}{(p + \alpha_2) \cosh [(p/\nu_2)^{1/2} b_2]} \qquad \text{(A15)}$$

Laplace and Hankel transforming the first equation of (25) and making use of the other three as well give

$$\frac{Q}{2\pi T_1 p} - a^2 s_1''(a, p) + \frac{K_2}{T_1} \frac{\partial s_2''(a, z, p)}{\partial z}\bigg|_{z=b_1} = \frac{p}{\nu_1} s_1''(a, p) \qquad \text{(A16)}$$

[*Hantush,* 1960, p. 3721], where $\nu_1 = T_1/S_1$.

If (A15) is differentiated with respect to z and the result evaluated at $z = b_1$, then, on combination with (A16), the following equation is obtained:

$$s_1''(a, p) = Q/2\pi T_1 p \cdot \left\{a^2 + \frac{p}{\nu_1} + \frac{K_2}{T_1}(p/\nu_2)^{1/2}\tanh\left[(p/\nu_2)^{1/2} b_2\right] + \frac{\gamma_2 p \operatorname{sech}^2\left[(p/\nu_2)^{1/2} b_2\right]}{p + \alpha_2 + \gamma_2(\nu_2 p)^{1/2}\tanh\left[(p/\nu_2)^{1/2} b_2\right]}\right\}^{-1} \qquad \text{(A17)}$$

Setting

$$\gamma = \left\{\frac{p}{\nu_1} + \frac{K_2}{T_1}\left(\frac{p}{\nu_2}\right)^{1/2} \tanh\left[\left(\frac{p}{\nu_2}\right)^{1/2} b_2\right] + \frac{\gamma_2 p \operatorname{sech}^2\left[(p/\nu_2)^{1/2} b_2\right]}{p + \alpha_2 + \gamma_2(\nu_2 p)^{1/2}\tanh\left[(p/\nu_2)^{1/2} b_2\right]}\right\}^{1/2} \qquad \text{(A18)}$$

and performing the Hankel inversion [*Hantush,* 1960, p. 3721] result in

$$s_1'(r, p) = \frac{Q}{2\pi T_1 p} K_0(\gamma r) = \frac{Q}{4\pi T_1 p}\int_0^\infty \frac{dy}{y} e^{-y-[(\gamma r)^2/4y]} \qquad \text{(A19)}$$

At this point, to facilitate the Laplace inversion, we will make asymptotic approximations of γ similar to those made by *Hantush* [1960, pp. 3722–3723]. The approximation that is valid for small time consists of considering values of p such that

$$(p/\nu_2)^{1/2} b_2 \geq 10^{1/2}$$

or

$$b_2/\nu_2^{1/2} \geq (10t)^{1/2}$$

the result being

$$\tanh\left[(p/\nu_2)^{1/2} b_2\right] \simeq 1$$

$$\operatorname{sech}^2\left[(p/\nu_2)^{1/2} b_2\right] \simeq 0$$

and

$$s_1'(r, p) = \frac{Q}{4\pi T_1 p}\int_0^\infty \frac{dy}{y} \cdot \exp\left\{-y - r^2[(p/\nu_1) + K_2(p/\nu_2)^{1/2}/T_1]/4y\right\} \qquad \text{(A20)}$$

Equation A20 was solved by *Hantush* [1960, pp. 3722–3723], and the solution is (26).

The large time approximation for γ uses p such that

$$(p/\nu_2)^{1/2} b_2 \leq 1/10^{1/2}$$

or

$$b_2/\nu_2^{1/2} \leq (t/10)^{1/2}$$

the result being

$$\tanh\left[(p/\nu_2)^{1/2} b_2\right] \simeq (p/\nu_2)^{1/2} b_2$$

and

$$\operatorname{sech}^2\left[(p/\nu_2)^{1/2} b_2\right] \simeq 1 - (p/\nu_2) b_2^2$$

With the above approximations (A19) becomes

$$s_r'(r, p) = \frac{Q}{2\pi T_1 p} K_0 \cdot \left[r\left(\frac{p}{A} + \frac{Cp}{p + a} - \frac{Bp^2}{p + a}\right)^{1/2}\right] = \frac{Q}{2\pi T_1 p} K_0\left[r\left(\frac{p(p + b)}{c(p + a)}\right)^{1/2}\right] \qquad \text{(A21)}$$

where

$$A = T_1/(S_2 + S_1)$$

$$B = \alpha_2 S_{y2} S_2 b_2/T_1(K_2 + \alpha_2 S_{y2} b_2)$$

$$C = \alpha_2 S_{y2} K_2/T_1(K_2 + \alpha_2 S_{y2} b_2)$$

$$a = \alpha_2 K_2/(K_2 + \alpha_2 S_{y2} b_2)$$

$$b = (a + CA)/(1 - BA)$$

$$c = A/(1 - BA)$$

Equation A2i was inverted by using methods of contour integration similar to those detailed in *Neuman and Witherspoon* [1969, pp. 21–40].

The first step in the Laplace inversion is evaluating the Laplace inverse of $K_0\{r[p(p+b)/c(p+a)]\}^{1/2}$ by using the Bromwich integral [*Boas,* 1966, pp. 613–614], which for this case becomes

$$L^{-1}K_0\left[r\left(\frac{p(p+b)}{c(p+a)}\right)^{1/2}\right]=\frac{1}{2\pi i}\cdot\int_{\xi-i\infty}^{\xi+i\infty}e^{zt}K_0\left[r\left(\frac{z(z+b)}{c(z+a)}\right)^{1/2}\right]dz \quad \text{(A22)}$$

where $\xi > \operatorname{Re} z$ and $z = x + iy$.

This integral is evaluated by solving the contour integral

$$\oint e^{zt}K_0\left[r\left(\frac{z(z+b)}{c(z+a)}\right)^{1/2}\right]dz = 0 \quad \text{(A23)}$$

over the contour shown in Figure 5 in the limit as $R \rightarrow \infty$. The convolution [*Boas,* 1966, pp. 607–610] of the Laplace inverse of $Q/2\pi T_1 p$ with the inverse evaluated above yields the final solution:

$$s_1=\frac{Q}{4\pi T_1}\left\{\int_0^a(1-e^{-ut})J_0\cdot\left[r\left(\frac{u(\eta a-u)}{c(a-u)}\right)^{1/2}\right]\frac{du}{u}+\int_{\eta a}^{\infty}(1-e^{-ut})J_0\left[r\left(\frac{u(\eta a-u)}{c(a-u)}\right)^{1/2}\right]\frac{du}{u}\right\} \quad \text{(A24)}$$

where $\eta = b/a$. Equation A24 is identical in form to that of *Boulton* [1955, p. 475].

We now follow *Boulton* [1963, p. 479] and transform (A24) by defining a new variable:

$$x^2 = u(\eta a - u)/\eta a(a - u) \quad \text{(A25)}$$

After some lengthy algebra we obtain

$$s_1=\frac{Q}{4\pi T_1}\int_0^{\infty}2\left[1-e^{-\mu_1}\left(\cosh\mu_2+\frac{at\eta(1-x^2)}{2\mu_2}\sinh\mu_2\right)\right]J_0\left(\frac{r}{\nu B}x\right)\frac{dx}{x} \quad \text{(A26)}$$

where $2\mu_1 = \eta at(1 + x^2)$, $2\mu_2 = at[\eta^2(1 + x^2)^2 - 4\eta x^2]^{1/2}$, and other variables are as defined earlier.

The final step consists of nondimensionalizing (A26) by using the definition

$$y^2 = \eta(r/B)^2(1 + x^2)/8\nu^2 \quad \text{(A27)}$$

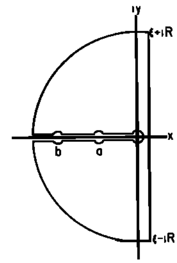


Fig. 5. Contour of integration used for the evaluation of the Bromwich integral.

and noting that

$$\frac{1}{u_y}=\frac{4a\nu^2t}{(r/B)^2}=\frac{4T_1t}{(S_1+S_2+S_{y2})r^2} \quad \text{(A28)}$$

which results in (27).

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